

Problem 3.15

If \$64 grows to \$128 in four years at a constant effective annual interest rate, how much will \$10,000 grow to in three years at the same rate of interest?

Solution

Let the constant effective annual interest rate be i . Since \$64 grows to \$128 in four years,

$$128 = 64(1 + i)^4.$$

Dividing by 64,

$$\frac{128}{64} = (1 + i)^4.$$

Therefore,

$$i = \sqrt[4]{\frac{128}{64}} - 1.$$

Hence,

$$i = \sqrt[4]{2} - 1 \approx 0.1892.$$

Now apply this rate to \$10,000 for three years:

$$X = 10000(1 + i)^3.$$

Substituting $i \approx 0.1892$,

$$X = 10000(1.1892)^3.$$

Therefore,

$$X \approx 16817.93$$

More accurately, since $(1 + i)^4 = 2$, we have $(1 + i) = 2^{1/4}$, so

$$X = 10000 (2^{1/4})^3 = 10000 2^{3/4} \approx 16817.93.$$